Problem Set 3

Instructor solutions

Counting, collisions, and derangements

100 points + 10 optional challenge points

Teaching note. This assignment consolidates discrete probability spaces and counting, then pushes the birthday and hat-check problems one step past the lectures: rigorous two-sided birthday bounds, collision patterns, the derangement recurrence, and the distribution of the number of correct hats. Allow approximately three and a half hours, excluding the challenge. Subpart scores appear in parentheses. Give credit for any correct model and reasoning; distinguish a wrong model from an arithmetic error carried through correctly.

Instructions. Identify the outcomes, their probabilities, and the event being counted. State whether order matters and repetition is allowed. Give exact answers except where decimal approximations are requested. A calculator or short program may be used for Problems 3, 6, and 7. Throughout, D_n is the number of derangements of n objects (permutations with no fixed point), and $p_n = D_n/n!$; you may use the formula $p_n = \sum_{k=0}^{n}(-1)^k/k!$ from class. By convention $D_0 = 1$.

I. Discrete probability spaces

1. A space with infinitely many outcomes [10 points]

Flip a fair coin until the first head appears and record the number K of flips used. Take $\Omega = \{1, 2, 3, \dots\}$, let $\mathcal{F} = 2^{\Omega}$, and give outcome k the mass 2^{-k}.

(a) Explain where the mass 2^{-k} comes from, and verify that the masses sum to one. What does this say about the possibility that no head ever appears? (3)

(b) Find $\mathbb{P}(K \text{ is even})$ and $\mathbb{P}(K \text{ is divisible by } 3)$. Find $\mathbb{P}(K > n)$ in two ways: by summing masses, and by describing the event in terms of the first n flips. (4)

(c) Show that there is *no* probability measure on $\Omega = \{1, 2, 3, \dots\}$ giving every outcome the same mass. Which axiom does the argument use that a finite space never needs? (3)

Solution

(a) $K = k$ means the flips are $T \cdots TH$ with $k-1$ tails: one specified string among 2^k equally likely strings of length k. The geometric series gives $\sum_{k\ge1} 2^{-k} = \frac{1/2}{1-1/2} = 1$. The outcomes "first head at flip k" already carry total probability one, so "no head ever" has probability zero; nothing is lost by omitting it from Ω.

(b) Both are geometric series:

$$\mathbb{P}(K \text{ even}) = \sum_{j\ge1} 4^{-j} = \frac{1/4}{1-1/4} = \frac{1}{3}, \qquad \mathbb{P}(3 \mid K) = \sum_{j\ge1} 8^{-j} = \frac{1/8}{1-1/8} = \frac{1}{7}.$$

Summing masses, $\mathbb{P}(K > n) = \sum_{k>n} 2^{-k} = 2^{-n}$. Directly, $K > n$ exactly when the first n flips are all tails, which has probability 2^{-n}.

(c) Suppose every outcome had mass c. Countable additivity applied to $\Omega = \bigcup_k \{k\}$ gives $1 = \mathbb{P}(\Omega) = \sum_{k\ge1} c$. If $c = 0$ the sum is 0; if $c > 0$ it diverges. Either way it is not 1. The argument uses *countable* additivity; on a finite space finite additivity suffices. (So "a uniformly random positive integer" is not a probability model.)

2. The larger of two dice [10 points]

Roll two fair dice, one red and one green, and let M be the larger of the two values shown (the common value if they agree).

(a) Give Ω and the mass of each outcome. Show that $\mathbb{P}(M \le k) = k^2/36$ for $k = 1, \dots, 6$, and deduce that $\mathbb{P}(M = k) = (2k-1)/36$. (4)

(b) Check that your six values of $\mathbb{P}(M = k)$ sum to one. Which familiar identity about odd numbers have you verified, and how does a 6×6 grid of outcomes explain it? (3)

(c) Recording only M produces a new probability space on $\{1, \dots, 6\}$. Find $\mathbb{P}(M \text{ is even})$ in that space. Compare with the probability that a single fair die is even, and explain the direction of the difference. (3)

Solution

(a) $\Omega = \{1, \dots, 6\}^2$ (red, green), each of the 36 ordered pairs having mass $1/36$. $M \le k$ means both dice show at most k: k^2 outcomes. Since $\{M \le k\}$ is the disjoint union of $\{M \le k-1\}$ and $\{M = k\}$,

$$\mathbb{P}(M = k) = \frac{k^2 - (k-1)^2}{36} = \frac{2k-1}{36}.$$

(b) $(1+3+5+7+9+11)/36 = 36/36$. This is the identity $1 + 3 + \cdots + (2n-1) = n^2$ at $n = 6$. In the grid, the outcomes with $M = k$ form an L-shaped border of $2k-1$ cells around the $(k-1) \times (k-1)$ square, and the six borders tile the whole grid.

(c) $\mathbb{P}(M \text{ even}) = (3+7+11)/36 = 21/36 = 7/12$, larger than $1/2$. The masses $(2k-1)/36$ increase with k, and each even value $2j$ outweighs the odd value $2j-1$ just below it, so the even values collect more than half of the mass. The recorded space has six outcomes, but they are not equally likely.

II. Permutations and combinations

3. Poker hands [12 points]

A hand is five cards dealt from a well-shuffled standard deck: 13 ranks in each of 4 suits. Give each probability as an exact fraction and as a decimal to four places.

(a) Describe a probability space in which a hand is an *unordered* set of cards, and count its outcomes. (2)

(b) Find the probability of a *full house*: three cards of one rank and two of another. (3)

(c) Find the probability of *exactly one pair*: two cards of one rank and three cards of three other, different ranks. (4)

(d) For *two pairs* (two cards of one rank, two of another, and a fifth card of a third rank) a student writes $13\binom{4}{2} \cdot 12\binom{4}{2} \cdot 44$. Explain the error, give the correct count, and explain why the same "first rank, then second rank" scheme was *not* an error in part (b). (3)

Solution

(a) Ω is the set of 5-element subsets of the deck, all equally likely, so $|\Omega| = \binom{52}{5} = 2{,}598{,}960$.

(b) Choose the rank of the triple and three of its four suits, then the rank of the pair and two of its suits: $13\binom{4}{3} \cdot 12\binom{4}{2} = 3744$. The probability is $3744/2{,}598{,}960 = 6/4165 \approx 0.0014$.

(c) Choose the paired rank and its two suits, then three other ranks as an unordered set, then a suit for each: $13\binom{4}{2}\binom{12}{3}4^3 = 1{,}098{,}240$, giving probability $352/833 \approx 0.4226$.

Solution (continued)

(d) The scheme orders the two paired ranks, but "kings and sevens" is the same hand as "sevens and kings": every hand is counted twice. The correct count is $\binom{13}{2}\binom{4}{2}^2 \cdot 44 = 123{,}552$, with probability $198/4165 \approx 0.0475$. In a full house the two ranks play *different* roles (triple and pair), so an ordered choice of ranks corresponds to exactly one hand. The rule: divide by the symmetry only when the roles are interchangeable.

4. Random arrangements of a word with repeats [12 points]

The eleven letter tiles of `MISSISSIPPI` (one M, four I's, four S's, two P's) are placed in a uniformly random order.

(a) How many distinguishable arrangements are there? Explain why all of them are equally likely, even though the underlying experiment permutes eleven physical tiles. (3)

(b) Find the probability that no two S's are adjacent. (Arrange the other letters first, then choose gaps.) (4)

(c) Find the probability that the first and last letters are the same, in two ways: by counting arrangements, and by looking only at the two physical tiles that land in the first and last positions. (5)

Solution

(a) There are $11!$ equally likely orderings of labeled tiles, and each distinguishable word arises from exactly $1!\,4!\,4!\,2!$ of them (permute identical letters among themselves). Equal-sized classes of equally likely outcomes are equally likely, and the number of words is $11!/(4!\,4!\,2!) = 34{,}650$.

(b) The seven non-S letters can be arranged in $7!/(4!\,2!) = 105$ ways. They create 8 gaps (including the two ends); the S's are non-adjacent exactly when they occupy four different gaps, $\binom{8}{4} = 70$ choices. Thus $105 \cdot 70 = 7350$ arrangements, and the probability is $7350/34{,}650 = 7/33$.

(c) *Counting words.* If both ends are I, the middle nine letters are M, II, SSSS, PP: $9!/(2!\,4!\,2!) = 3780$ words. Both ends S gives 3780 by symmetry, and both ends P gives $9!/(4!\,4!) = 630$. The probability is $(3780 + 3780 + 630)/34{,}650 = 8190/34{,}650 = 13/55$.

Two tiles. The ordered pair of tiles in positions 1 and 11 is uniform over the $11 \cdot 10$ ordered pairs of distinct tiles. The number of pairs with equal letters is $4 \cdot 3 + 4 \cdot 3 + 2 \cdot 1 = 26$, so the probability is $26/110 = 13/55$.

5. Committees and Vandermonde's identity [10 points]

A five-person committee is chosen uniformly at random from a club of 6 women and 9 men.

(a) Find the probability that the committee has exactly two women. (3)

(b) Find the probability that it includes at least one woman and at least one man. (3)

(c) Prove *Vandermonde's identity* $\displaystyle\sum_{j=0}^{k} \binom{m}{j}\binom{n}{k-j} = \binom{m+n}{k}$ by a counting argument (with $\binom{a}{b} = 0$ when $b > a$). What does it say about the probabilities in part (a), taken over all possible numbers of women? (4)

Solution

(a) There are $\binom{15}{5} = 3003$ equally likely committees, of which $\binom{6}{2}\binom{9}{3} = 15 \cdot 84 = 1260$ have exactly two women. The probability is $1260/3003 = 60/143$.

(b) The complement is "all men or all women," two disjoint events with $\binom{9}{5} = 126$ and $\binom{6}{5} = 6$ committees. The probability is $1 - 132/3003 = 1 - 4/91 = 87/91$.

Solution (continued)

(c) Count the k-person committees from m women and n men. Directly there are $\binom{m+n}{k}$. Alternatively classify by the number j of women: choose them in $\binom{m}{j}$ ways and the $k-j$ men in $\binom{n}{k-j}$ ways. The classes are disjoint and exhaustive, so the sum equals $\binom{m+n}{k}$. Dividing by $\binom{m+n}{k}$: the probabilities $\binom{m}{j}\binom{n}{k-j}/\binom{m+n}{k}$ of "exactly j women," $j = 0, \ldots, k$, sum to one, as the probabilities of a partition of Ω must.

III. Birthday collisions

6. Two honest bounds for the birthday problem [12 points]

Each of n customers independently chooses a four-digit PIN uniformly from the $d = 10^4$ possibilities. Let N_n be the event that all n PINs are different, so that $\mathbb{P}(N_n) = \prod_{k=1}^{n-1}(1 - k/d)$. In class we *approximated* $\ln(1-x) \approx -x$. Here you will replace the approximation by inequalities.

(a) Prove that $1 - x \le e^{-x}$ for every real x. (3)

(b) Deduce that $\mathbb{P}(\text{some two PINs agree}) \ge 1 - \exp(-n(n-1)/(2d))$: the approximation from class is a rigorous *lower* bound for the collision probability. (3)

(c) Let E_{ij} be the event that customers i and j choose the same PIN. Using $\mathbb{P}(A \cup B) \le \mathbb{P}(A) + \mathbb{P}(B)$, show that $\mathbb{P}(\text{some two PINs agree}) \le \binom{n}{2}/d$. (3)

(d) Use (c) to show that a collision is less likely than not when $n \le 100$, and (b) to show that it is more likely than not when $n \ge 119$ (use $\ln 2 \approx 0.693147$). Then compute the exact collision probabilities for $n = 118$ and $n = 119$ to four decimal places. (3)

Solution

(a) Let $g(x) = e^{-x} - (1 - x)$. Then $g'(x) = 1 - e^{-x}$ is negative for $x < 0$ and positive for $x > 0$, so g has its minimum at $x = 0$, where $g(0) = 0$. Hence $g \ge 0$. (Equivalently: the convex function e^{-x} lies above its tangent line at 0.)

(b) Every factor satisfies $0 \le 1 - k/d \le e^{-k/d}$, and inequalities between nonnegative numbers may be multiplied:

$$\mathbb{P}(N_n) \le \prod_{k=1}^{n-1} e^{-k/d} = \exp\Big(-\frac{1}{d}\sum_{k=1}^{n-1} k\Big) = \exp\Big(-\frac{n(n-1)}{2d}\Big).$$

Take complements.

(c) The collision event is $\bigcup_{i<j} E_{ij}$, a union of $\binom{n}{2}$ events. By induction the two-event inequality gives $\mathbb{P}(\bigcup_m A_m) \le \sum_m \mathbb{P}(A_m)$ for finitely many events. Whatever PIN customer i chose, customer j matches it with probability $1/d$, so $\mathbb{P}(E_{ij}) = 1/d$ and the bound is $\binom{n}{2}/d$.

(d) For $n \le 100$ the collision probability is at most $\binom{100}{2}/10^4 = 0.495 < 1/2$; it is nondecreasing in n, so the single case $n = 100$ suffices. By (b), the collision probability exceeds $1/2$ as soon as $n(n-1) > 2d \ln 2 \approx 13{,}862.9$; since $118 \cdot 117 = 13{,}806$ and $119 \cdot 118 = 14{,}042$, this holds for $n \ge 119$. Exactly,

$$1 - \mathbb{P}(N_{118}) \approx 0.4999, \qquad 1 - \mathbb{P}(N_{119}) \approx 0.5058,$$

so the threshold is $n = 119$. (To more places $1 - \mathbb{P}(N_{118}) = 0.499936$, a remarkably near miss. The lower bound from (b) is 0.4986 and 0.5045 at these two values: accurate to about one part in a thousand.)

7. What kind of collision? [12 points]

(a) Four labeled people have independent birth *months*, uniform on 12 months. Classify the 12^4 outcomes by pattern: all four different; exactly one shared pair (the other two different from it and from each other); two different shared pairs; three alike and one different; all four alike. Count each pattern, and check that the counts total 12^4. (6)

(b) Now n people have independent birthdays, uniform on d days, with $2 \le n \le d+1$. Show that the probability that exactly one pair shares a birthday and all other birthdays are different from it and from each other is $\binom{n}{2}\, d(d-1)\cdots(d-n+2)/d^n$. (3)

(c) For $n = 23$ and $d = 365$, compute to four decimal places the probabilities of no shared birthday, of the pattern in (b), and of any more complicated coincidence. (3)

Solution

(a) In each case choose which people are grouped, then assign distinct months to the groups.

Pattern	Count	
All different	$12 \cdot 11 \cdot 10 \cdot 9$	11,880
One pair	$\binom{4}{2} \cdot 12 \cdot 11 \cdot 10$	7,920
Two pairs	$3 \cdot 12 \cdot 11$	396
Three and one	$4 \cdot 12 \cdot 11$	528
All alike	12	12
Total		$20{,}736 = 12^4$

For two pairs, there are 3 ways to split four people into two unordered pairs; once the split is fixed, the pair containing person 1 is distinguished, so its month and the other pair's month form an ordered choice, $12 \cdot 11$. (The probabilities are 55/96, 55/144, 11/576, 11/432, 1/1728.)

(b) Choose the pair in $\binom{n}{2}$ ways. The pair together with the $n-2$ other people forms $n-1$ labeled groups needing distinct days: $d(d-1)\cdots(d-n+2)$ assignments. Different choices give different birthday lists, and all d^n lists are equally likely.

(c) No match: 0.4927. Exactly one pair: 0.3634. Anything more complicated (two or more pairs, a triple, ...): $1 - 0.4927 - 0.3634 = 0.1439$. So when a class of 23 has a coincidence, it is a single pair about 72% of the time.

IV. Derangements

8. A recurrence for derangements [10 points]

(a) Prove that $D_n = (n-1)\,(D_{n-1} + D_{n-2})$ for $n \ge 2$, by a counting argument. Suggested route: in a derangement guest 1 receives some hat $j \ne 1$; separate the cases in which guest j does, or does not, receive hat 1. (5)

(b) Starting from $D_0 = 1$ and $D_1 = 0$, compute $D_2, \ldots, D_6$, and check D_6 against the formula for p_6. (2)

(c) The alternating-series estimate gives $|p_n - 1/e| < 1/(n+1)!$. Deduce that for $n \ge 2$, D_n is the integer nearest to $n!/e$, and confirm this for $n = 6$. (3)

Solution

(a) There are $n-1$ choices for the hat j received by guest 1; by symmetry each contributes the same number of derangements. *Case 1: guest j receives hat 1.* Guests 1 and j have swapped, and the remaining $n-2$

Solution (continued)

guests must derange their own $n-2$ hats: D_{n-2} ways. *Case 2: guest j does not receive hat 1.* The guests other than 1 must receive the hats other than j, with guest j forbidden hat 1 and every other guest i forbidden hat i. Renaming hat 1 as "hat j" makes this exactly a derangement of $n-1$ guests: D_{n-1} ways. The cases are disjoint and exhaustive, so $D_n = (n-1)(D_{n-2} + D_{n-1})$.

(b) $D_2 = 1 \cdot (0+1) = 1$, $D_3 = 2(1+0) = 2$, $D_4 = 3(2+1) = 9$, $D_5 = 4(9+2) = 44$, $D_6 = 5(44+9) = 265$. The formula gives $p_6 = \frac{1}{2} - \frac{1}{6} + \frac{1}{24} - \frac{1}{120} + \frac{1}{720} = \frac{360-120+30-6+1}{720} = \frac{265}{720}$.

(c) Multiply by $n!$: $|D_n - n!/e| < n!/(n+1)! = 1/(n+1) \le 1/3$ for $n \ge 2$. An integer within $1/2$ of a real number is the nearest integer to it, and it is unique. For $n = 6$: $720/e \approx 264.87$, nearest integer 265.

9. How many guests get the right hat? [12 points]

The hats of n guests are returned as a uniformly random permutation.

(a) Show that the probability that *exactly* k guests receive their own hats is $\binom{n}{k} D_{n-k}/n! = p_{n-k}/k!$. (3)

(b) For $n = 5$, tabulate the number of permutations with exactly k correct hats for $k = 0, 1, \ldots, 5$, and check the total. (3)

(c) Explain in words why exactly $n-1$ correct hats is impossible, and how the formula in (a) knows this. For fixed k, find the limit of the probability in (a) as $n \to \infty$. (3)

(d) Show that the total number of correct hats, summed over all $n!$ permutations, is exactly $n!$, by counting pairs (permutation, guest with the correct hat) in a second way. Verify this with your table from (b). What is the average number of correct hats, and how does it depend on n? (3)

Solution

(a) Choose which k guests are correct, $\binom{n}{k}$ ways; the other $n-k$ guests must then derange their own hats, D_{n-k} ways. Different choices give different permutations. Dividing by $n!$,

$$\frac{n!}{k!\,(n-k)!} \cdot \frac{D_{n-k}}{n!} = \frac{1}{k!} \cdot \frac{D_{n-k}}{(n-k)!} = \frac{p_{n-k}}{k!}.$$

(b) Using $D_0, \ldots, D_5 = 1, 0, 1, 2, 9, 44$:

k	0	1	2	3	4	5	Total
$\binom{5}{k} D_{5-k}$	44	45	20	10	0	1	120

(c) If $n-1$ guests have their own hats, the one remaining hat belongs to the one remaining guest, who is then also correct. In the formula this is $D_1 = 0$. For fixed k, $p_{n-k} \to 1/e$, so the probability tends to $e^{-1}/k!$. (These limits sum to $e^{-1} \sum_k 1/k! = 1$: they form a probability distribution on $\{0, 1, 2, \ldots\}$, which we shall meet again as the Poisson distribution with mean 1.)

(d) Count pairs (σ, i) with $\sigma(i) = i$. Grouping by σ gives the total number of correct hats. Grouping by i: the permutations fixing i are the $(n-1)!$ permutations of the other hats, so there are $n \cdot (n-1)! = n!$ pairs. Table check: $0 \cdot 44 + 1 \cdot 45 + 2 \cdot 20 + 3 \cdot 10 + 4 \cdot 0 + 5 \cdot 1 = 120$. The average number of correct hats is $n!/n! = 1$ for every n: a larger party offers more chances, each proportionally less likely.

Optional challenge: Secret Santa [10 extra points]

In a Secret Santa, $n \geq 3$ people draw names from a hat, and the draw is repeated from scratch until nobody has drawn their own name.

(a) Explain why the final assignment is uniformly distributed over the D_n derangements. For $n = 5$, find the probability that more than three draws are needed. (3)

(b) Find the probability that persons 1 and 2 draw each other, in terms of derangement numbers, and evaluate it for $n = 6$. (3)

(c) Find the probability that person 1 draws person 2, for general n. Then use (b) to find the conditional probability that 2 drew 1, given that 1 drew 2, for $n = 6$. Is reciprocation more or less likely than the unconditional probability that 2 draws 1? (4)

Solution

(a) Each draw is a uniform permutation, and draws are independent. The first successful draw is a uniform permutation conditioned on being a derangement; conditioning a uniform distribution on a subset gives the uniform distribution on that subset, since every derangement σ has conditional probability $(1/n!)/(D_n/n!) = 1/D_n$. For $n = 5$ a draw fails with probability $1 - 44/120 = 19/30$, so more than three draws are needed with probability $(19/30)^3 = 6859/27{,}000 \approx 0.254$.

(b) If 1 and 2 swap, the other $n - 2$ people must derange among themselves: D_{n-2} of the D_n equally likely derangements. For $n = 6$ this is $D_4/D_6 = 9/265$.

(c) Relabeling persons $2, \ldots, n$ maps derangements to derangements, so person 1 is equally likely to draw each of the other $n - 1$ people: the probability is $1/(n-1)$. For $n = 6$,

$$\mathbb{P}(2 \to 1 \mid 1 \to 2) = \frac{9/265}{1/5} = \frac{9}{53} \approx 0.170 < \frac{1}{5}.$$

Reciprocation is slightly *less* likely than the unconditional $1/5$. In general, by the recurrence of Problem 8 the derangements with $1 \to 2$ number $D_{n-1} + D_{n-2}$, of which the D_{n-2} of Case 1 reciprocate. The formula for p_m gives $D_m = mD_{m-1} + (-1)^m$, and therefore

$$\mathbb{P}(2 \to 1 \mid 1 \to 2) = \frac{D_{n-2}}{D_{n-1} + D_{n-2}} = \frac{1}{n + (-1)^{n-1}/D_{n-2}},$$

which is close to $1/n$ rather than $1/(n-1)$; at $n = 6$ it is $1/(6 - \frac{1}{9}) = 9/53$. It is below $1/(n-1)$ for every $n \geq 3$ except $n = 4$, where the two agree.

Suggested credit: as marked, (3), (3), (4); in (c) award 2 for $1/(n-1)$ with a symmetry argument and 2 for the conditional probability and comparison.